

Automation and Sectoral Reallocation

Online Appendix

Dennis Hutschenreiter¹

Tommaso Santini²

Eugenia Vella³

February 11, 2021

¹Universitat Autònoma de Barcelona and Barcelona GSE. *e-mail:* Dennis.Hutschenreiter@uab.cat

²Universitat Autònoma de Barcelona and Barcelona GSE. *e-mail:* tommaso.santini@uab.cat.

³Athens University of Economics and Business, Greek and European Economy Observatory at ELIAMEP (Hellenic Foundation for European and Foreign Policy), and Fundació MOVE (Markets, Organizations and Votes in Economics). *e-mail:* evella@aueb.gr

1 The maximization problem of the household

The Lagrangian is given by:

$$\begin{aligned} \mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \Bigg\{ & \frac{c_t^{1-\eta}}{1-\eta} + \Phi \frac{[1 - u_t - n_t^M - n_t^S]^{1-\varphi}}{1-\varphi} \\ & - \lambda_{ct} [c_t + k_{t+1} - (1-\delta)k_t - r_t k_t - w_t^M n_t^M - w_t^S n_t^S - \bar{b}_t u_t - \Pi_t^M - \Pi_t^S] \\ & - \lambda_{n^M t} [n_{t+1}^M - (1-\sigma^M) n_t^M - \psi_t^{hM} s_t u_t] \\ & - \lambda_{n^S t} [n_{t+1}^S - (1-\sigma^S) n_t^S - \psi_t^{hS} (1-s_t) u_t] \Bigg\}. \end{aligned}$$

The first-order conditions with respect to c_t , k_{t+1} , n_{t+1}^M , n_{t+1}^S , u_t , and s_t are,
[wrt c_t]

$$c_t^{-\eta} - \lambda_{ct} = 0, \quad (\text{A.1})$$

[wrt k_{t+1}]

$$-\beta^t \lambda_{ct} + \beta^{t+1} E_t [\lambda_{ct+1} (1-\delta + r_{t+1})] = 0, \quad (\text{A.2})$$

[wrt n_{t+1}^M]

$$-\beta^t \lambda_{n^M t} + \beta^{t+1} E_t [-\Phi l_{t+1}^{-\varphi} + \lambda_{ct+1} w_{t+1}^M + \lambda_{n^M t+1} (1-\sigma^M)] = 0, \quad (\text{A.3})$$

[wrt n_{t+1}^S]

$$-\beta^t \lambda_{n^S t} + \beta^{t+1} E_t [-\Phi l_{t+1}^{-\varphi} + \lambda_{ct+1} w_{t+1}^S + \lambda_{n^S t+1} (1-\sigma^S)] = 0, \quad (\text{A.4})$$

[wrt u_t]

$$-\beta^t \Phi l_t^{-\varphi} + \beta^t \lambda_{ct} \bar{b}_t + \beta^t \lambda_{n^M t} \psi_t^{hM} s_t + \beta^t \lambda_{n^S t} \psi_t^{hS} (1-s_t) = 0, \quad (\text{A.5})$$

[wrt s_t]

$$\lambda_{n^M t} \psi_t^{hM} - \lambda_{n^S t} \psi_t^{hS} = 0. \quad (\text{A.6})$$

To simplify the system of equations, we combine (A.1) and (A.2),

$$c_t^{-\eta} = \beta E_t [c_{t+1}^{-\eta} (1-\delta + r_{t+1})], \quad (\text{A.7})$$

and then rearrange terms in (A.3) - (A.6),

$$\lambda_{n^M t} = \beta E_t [-\Phi l_{t+1}^{-\varphi} + c_{t+1}^{-\eta} w_{t+1}^M + \lambda_{n^M t+1} (1-\sigma^M)], \quad (\text{A.8})$$

$$\lambda_{n^S t} = \beta E_t \left[-\Phi l_{t+1}^{-\varphi} + c_{t+1}^{-\eta} w_{t+1}^S + \lambda_{n^S t+1} (1 - \sigma^S) \right], \quad (\text{A.9})$$

$$\Phi l_t^{-\varphi} = \lambda_{n^M t} \psi_t^{hM} s_t + \lambda_{n^S t} \psi_t^{hS} (1 - s_t) + \lambda_{ct} \bar{b}_t, \quad (\text{A.10})$$

$$\lambda_{n_t^M} \psi_t^{hM} = \lambda_{n_t^S} \psi_t^{hS}. \quad (\text{A.11})$$

2 The wage bargaining problem

2.1 Manufacturing sector

The maximization problem is written as,

$$\max_{w_t^M} \left\{ (1 - \vartheta^M) \ln V_{n^M t}^h + \vartheta^M \ln V_{n^M t}^f \right\}, \quad (\text{A.12})$$

where

$$V_{n^M t}^h = -\Phi l_t^{-\varphi} + \lambda_{ct} w_t^M + (1 - \sigma^M) \lambda_{n^M t}, \quad (\text{A.13})$$

and

$$V_{n^M t}^f = p_t^M (1 - \zeta) \left(\frac{M_t}{n_t^M} \right)^{\frac{1}{\alpha}} - w_t^M + \frac{(1 - \sigma^M) \kappa^M}{\psi_t^{fM}}. \quad (\text{A.14})$$

The first-order condition is written as,

$$\vartheta^M V_{n^M t}^h = (1 - \vartheta^M) \lambda_{ct} V_{n^M t}^f.$$

Using (A.13) and (A.14), we obtain,

$$\begin{aligned} \vartheta^M (\lambda_{ct} w_t^M - \Phi l_t^{-\varphi} + (1 - \sigma^M) \lambda_{n^M t}) = \\ (1 - \vartheta^M) \lambda_{ct} \left(p_t^M (1 - \zeta) \left(\frac{M_t}{n_t^M} \right)^{\frac{1}{\alpha}} - w_t^M + \frac{(1 - \sigma^M) \kappa^M}{\psi_t^{fM}} \right), \end{aligned}$$

$$\begin{aligned} \vartheta^M \lambda_{ct} w_t^M + \vartheta^M (-\Phi l_t^{-\varphi} + (1 - \sigma^M) \lambda_{n^M t}) = \\ - (1 - \vartheta^M) \lambda_{ct} w_t^M + (1 - \vartheta^M) \lambda_{ct} \left(p_t^M (1 - \zeta) \left(\frac{M_t}{n_t^M} \right)^{\frac{1}{\alpha}} + \frac{(1 - \sigma^M) \kappa^M}{\psi_t^{fM}} \right), \end{aligned}$$

$$\vartheta^M \lambda_{ct} w_t^M + (1 - \vartheta^M) \lambda_{ct} w_t^M =$$

$$-\vartheta^M (-\Phi l_t^{-\varphi} + (1 - \sigma^M) \lambda_{n^M t}) + (1 - \vartheta^M) \lambda_{ct} \left(p_t^M (1 - \zeta) \left(\frac{M_t}{n_t^M} \right)^{\frac{1}{\alpha}} + \frac{(1 - \sigma^M) \kappa^M}{\psi_t^{fM}} \right).$$

The final expression for the wage in the manufacturing sector is,

$$w_t^M = (1 - \vartheta^M) \left(p_t^M (1 - \zeta) \left(\frac{M_t}{n_t^M} \right)^{\frac{1}{\alpha}} + \frac{(1 - \sigma^M) \kappa^M}{\psi_t^{fM}} \right) - \frac{\vartheta^M}{\lambda_{ct}} ((1 - \sigma^M) \lambda_{n^M t} - \Phi l_t^{-\varphi}).$$

2.2 Service sector

Similarly, the maximization problem is written as,

$$\max_{w_t^S} \left\{ (1 - \vartheta^S) \ln V_{n^S t}^h + \vartheta^S \ln V_{n^S t}^f \right\}, \quad (\text{A.15})$$

where

$$V_{n^S t}^h = \lambda_{ct} w_t^S - \Phi l_t^{-\varphi} + (1 - \sigma^S) \lambda_{n^S t}, \quad (\text{A.16})$$

and

$$V_{n^S t}^f = p_t^S b \frac{S_t}{n_t^S} - w_t^S + \frac{(1 - \sigma^S) \kappa^S}{\psi_t^{fS}}. \quad (\text{A.17})$$

The first-order condition is written as,

$$\vartheta^S V_{n^S t}^h = (1 - \vartheta^S) \lambda_{ct} V_{n^S t}^f.$$

Using (A.16) and (A.17) we obtain,

$$\vartheta^S (\lambda_{ct} w_t^S - \Phi l_t^{-\varphi} + (1 - \sigma^S) \lambda_{n^S t}) = (1 - \vartheta^S) \lambda_{ct} (p_t^S b \frac{S_t}{n_t^S} - w_t^S + \frac{(1 - \sigma^S) \kappa^S}{\psi_t^{fS}}).$$

We perform a simple algebra below,

$$\vartheta^S \lambda_{ct} w_t^S + \vartheta^S (-\Phi l_t^{-\varphi} + (1 - \sigma^S) \lambda_{n^S t}) = -(1 - \vartheta^S) \lambda_{ct} w_t^S + (1 - \vartheta^S) \lambda_{ct} (p_t^S b \frac{S_t}{n_t^S} + \frac{(1 - \sigma^S) \kappa^S}{\psi_t^{fS}}),$$

$$\vartheta^S \lambda_{ct} w_t^S + (1 - \vartheta^S) \lambda_{ct} w_t^S = (1 - \vartheta^S) \lambda_{ct} \left(p_t^S b \frac{S_t}{n_t^S} + \frac{(1 - \sigma^S) \kappa^S}{\psi_t^{fS}} \right) - \vartheta^S (-\Phi l_t^{-\varphi} + (1 - \sigma^S) \lambda_{n^{S_t}}),$$

$$w_t^S \lambda_{ct} = (1 - \vartheta^S) \lambda_{ct} \left(p_t^S b \frac{S_t}{n_t^S} + \frac{(1 - \sigma^S) \kappa^S}{\psi_t^{fS}} \right) - \vartheta^S (-\Phi l_t^{-\varphi} + (1 - \sigma^S) \lambda_{n^{S_t}}).$$

The final expression for the equilibrium wage is given by,

$$w_t^S = (1 - \vartheta^S) \left(p_t^S b \frac{S_t}{n_t^S} + \frac{(1 - \sigma^S) \kappa^S}{\psi_t^{fS}} \right) - \frac{\vartheta^S}{\lambda_{ct}} ((1 - \sigma^S) \lambda_{n^{S_t}} - \Phi l_t^{-\varphi}).$$

3 Resource constraint

From the household budget constraint,

$$c_t + i_t = r_t k_t + w_t^M n_t^M + w_t^S n_t^S + \bar{b}_t u_t - T_t + \Pi_t^M + \Pi_t^S.$$

We substitute the expressions of the instantaneous profits of firms,

$$\Pi_t^M = p_t^M M_t - w_t^M n_t^M - r_t k_t - \kappa^M v_t^M,$$

$$\Pi_t^S = p_t^S S_t - w_t^S n_t^S - \kappa^S v_t^S,$$

and obtain,

$$c_t + i_t = p_t^M M_t + p_t^S S_t - \kappa^S v_t^S - \kappa^M v_t^M.$$

Because of the constant returns to scale and frictionless production of the final good, we have,

$$Y_t = p_t^M M_t + p_t^S S_t.$$

So, we obtain the resource constraint,

$$Y_t = c_t + i_t + \kappa^S v_t^S + \kappa^M v_t^M.$$

4 Model Solution

We report below all equations that need to be satisfied in equilibrium.

Labor markets

$$m_t^M = \mu_1(v_t^M)^{\mu_2}(u_t^M)^{1-\mu_2} \quad (\text{E.1})$$

$$m_t^S = \mu_1(v_t^S)^{\mu_2}(u_t^S)^{1-\mu_2} \quad (\text{E.2})$$

$$n_{t+1}^M = (1 - \sigma^M)n_t^M + m_t^M \quad (\text{E.3})$$

$$n_{t+1}^S = (1 - \sigma^S)n_t^S + m_t^S \quad (\text{E.4})$$

$$\psi_t^{hM} \equiv \frac{m_t^M}{u_t^M} \quad (\text{E.5})$$

$$\psi_t^{fM} \equiv \frac{m_t^M}{v_t^M} \quad (\text{E.6})$$

$$\psi_t^{hS} \equiv \frac{m_t^S}{u_t^S} \quad (\text{E.7})$$

$$\psi_t^{fS} \equiv \frac{m_t^S}{v_t^S} \quad (\text{E.8})$$

Household

$$c_t^{-\eta} = \beta E_t [c_{t+1}^{-\eta}(1 - \delta + r_{t+1})] \quad (\text{E.9})$$

$$\lambda_{n^M_t} = \beta E_t [-\Phi l_{t+1}^{-\varphi} + c_{t+1}^{-\eta} w_{t+1}^M + \lambda_{n^M_{t+1}}(1 - \sigma^M)] \quad (\text{E.10})$$

$$\lambda_{n^S_t} = \beta E_t [-\Phi l_{t+1}^{-\varphi} + c_{t+1}^{-\eta} w_{t+1}^S + \lambda_{n^S_{t+1}}(1 - \sigma^S)] \quad (\text{E.11})$$

$$\Phi l_t^{-\varphi} = \lambda_{n^M_t} \psi_t^{hM} s_t + \lambda_{n^S_t} \psi_t^{hS} (1 - s_t) + \lambda_{ct} \bar{b}_t \quad (\text{E.12})$$

$$\lambda_{n^M_t} \psi_t^{hM} = \lambda_{n^S_t} \psi_t^{hS} \quad (\text{E.13})$$

$$n_t^M + n_t^S + u_t + l_t = 1 \quad (\text{E.14})$$

$$u_t^M = s_t u_t \quad (\text{E.15})$$

$$u_t^S = (1 - s_t) u_t \quad (\text{E.16})$$

$$c_t + k_{t+1} = (1 - \delta + r_t) k_t + w_t^M n_t^M + w_t^S n_t^S + \bar{b}_t u_t - T_t + \Pi_t^M + \Pi_t^S \quad (\text{E.17})$$

Firms

$$\Lambda_{t,t+1} = \beta \left(\frac{\lambda_{n^{c_{t+1}}}}{\lambda_{n^{c_t}}} \right)^{-\eta} \quad (\text{E.18})$$

$$\frac{\kappa_t^M}{\psi_t^{fM}} = E_t \Lambda_{t,t+1} \left[p_{t+1}^M (1 - \zeta) \left(\frac{M_{t+1}}{n_{t+1}^M} \right)^{\frac{1}{\alpha}} - w_{t+1}^M + \frac{(1 - \sigma^M) \kappa_t^M}{\psi_{t+1}^{fM}} \right] \quad (\text{E.19})$$

$$r_t = p_t^M \cdot \zeta \left(\frac{M_t}{k_t} \right)^{\frac{1}{\alpha}} \quad (\text{E.20})$$

$$p_t^M = \gamma \left(\frac{Y_t}{M_t} \right)^{\frac{1}{\chi}} \quad (\text{E.21})$$

$$M_t = \left[\zeta k_t^{\frac{\alpha-1}{\alpha}} + (1 - \zeta) (n_t^M)^{\frac{\alpha-1}{\alpha}} \right]^{\frac{\alpha}{\alpha-1}}, \quad (\text{E.22})$$

$$Y_t = \left[\gamma M_t^{\frac{\chi-1}{\chi}} + (1 - \gamma) S_t^{\frac{\chi-1}{\chi}} \right]^{\frac{\chi}{\chi-1}}, \quad (\text{E.23})$$

$$\frac{\kappa_t^S}{\psi_t^{fS}} = E_t \Lambda_{t,t+1} \left[p_{t+1}^S b \frac{S_{t+1}}{n_{t+1}^S} - w_{t+1}^S + \frac{(1 - \sigma^S) \kappa_t^S}{\psi_{t+1}^{fS}} \right] \quad (\text{E.24})$$

$$p_t^S = (1 - \gamma) \left(\frac{Y_t}{S_t} \right)^{\frac{1}{\chi}} \quad (\text{E.25})$$

$$S_t = B (n_t^S)^b \quad (\text{E.26})$$

Wages

$$w_t^S = (1 - \vartheta^S) \left(p_t^S b \frac{S_t}{n_t^S} + \frac{(1 - \sigma^S) \kappa^S}{\psi_t^{fS}} \right) + \frac{\vartheta^S}{\lambda_{ct}} (\Phi l_t^{-\varphi} - (1 - \sigma^S) \lambda_{n^S t}) \quad (\text{E.27})$$

$$w_t^M = (1 - \vartheta^M) \left(p_t^M (1 - \zeta) \left(\frac{M_t}{n_t^M} \right)^{\frac{1}{\alpha}} + \frac{(1 - \sigma^M) \kappa^M}{\psi_t^{fM}} \right) + \frac{\vartheta^M}{\lambda_{ct}} (\Phi l_t^{-\varphi} - (1 - \sigma^M) \lambda_{n^M t}) \quad (\text{E.28})$$

5 Steady-state system of equations

$$m^M = \mu_1 (v^M)^{\mu_2} (u^M)^{1-\mu_2} \quad (\text{S.1})$$

$$m^S = \mu_1 (v^S)^{\mu_2} (u^S)^{1-\mu_2} \quad (\text{S.2})$$

$$n^M = \frac{m^M}{\sigma^M} \quad (\text{S.3})$$

$$n^S = \frac{m^S}{\sigma^S} \quad (\text{S.4})$$

$$\psi^{hM} \equiv \frac{m^M}{u^M}, \quad (\text{S.5})$$

$$\psi^{fM} \equiv \frac{m^M}{v^M}, \quad (\text{S.6})$$

$$\psi^{hS} \equiv \frac{m^S}{u^S}, \quad (\text{S.7})$$

$$\psi^{fS} \equiv \frac{m^S}{v^S} \quad (\text{S.8})$$

Household

$$r = \frac{1 - \beta}{\beta} + \delta \quad (\text{S.9})$$

$$\lambda_{n^M} = \beta [-\Phi l^{-\varphi} + c^{-\eta} w^M + \lambda_{n^M} (1 - \sigma^M)] \quad (\text{S.10})$$

$$\lambda_{n^S} = \beta [-\Phi l^{-\varphi} + c^{-\eta} w^S + \lambda_{n^S} (1 - \sigma^S)] \quad (\text{S.11})$$

$$\Phi l^{-\varphi} = \lambda_{n^M} \psi^{hM} s + \lambda_{n^S} \psi^{hS} (1 - s) + \lambda_c \bar{b} \quad (\text{S.12})$$

$$\lambda_{n^M} \psi^{hM} = \lambda_{n^S} \psi^{hS} \quad (\text{S.13})$$

$$n^M + n^S + u + l = 1 \quad (\text{S.14})$$

$$u^M = su \quad (\text{S.15})$$

$$u^S = (1 - s)u \quad (\text{S.16})$$

$$c = k(r - \delta) + w^M n^M + w^S n^S + \bar{b}u \quad (\text{S.17})$$

Firms

$$\frac{\kappa^S}{\psi^{fS}} = \beta \left[p^S b \frac{S}{n^S} - w^S + \frac{(1 - \sigma^S) \kappa^S}{\psi^{fS}} \right] \quad (\text{S.18})$$

$$p^S = (1 - \gamma) \left(\frac{Y}{S} \right)^{\frac{1}{\chi}} \quad (\text{S.19})$$

$$S = B(n^S)^b \quad (\text{S.20})$$

$$\frac{\kappa^M}{\psi_t^{fM}} = \beta \left[p^M (1 - \zeta) \left(\frac{M}{n^M} \right)^{\frac{1}{\alpha}} - w^M + \frac{(1 - \sigma^M) \kappa^M}{\psi^{fM}} \right] \quad (\text{S.21})$$

$$r = p^M \cdot \zeta \left(\frac{M}{k} \right)^{\frac{1}{\alpha}} \quad (\text{S.22})$$

$$p^M = \gamma \left(\frac{Y}{M} \right)^{\frac{1}{\chi}} \quad (\text{S.23})$$

$$M = \left[\zeta k^{\frac{\alpha-1}{\alpha}} + (1 - \zeta) (n^M)^{\frac{\alpha-1}{\alpha}} \right]^{\frac{\alpha}{\alpha-1}}, \quad (\text{S.24})$$

Wages

$$w^S = (1 - \vartheta^S) \left(p^S b \frac{S}{n^S} + \frac{(1 - \sigma^S) \kappa^S}{\psi^{fS}} \right) + \frac{\vartheta^S}{\lambda_c} (\Phi l^{-\varphi} - (1 - \sigma^S) \lambda_{n^S}) \quad (\text{S.25})$$

$$w^M = (1 - \vartheta^M) \left(p^M (1 - \zeta) \left(\frac{M}{n^M} \right)^{\frac{1}{\alpha}} + \frac{(1 - \sigma^M) \kappa^M}{\psi^{fM}} \right) + \frac{\vartheta^M}{\lambda_c} (\Phi l^{-\varphi} - (1 - \sigma^M) \lambda_{n^M}) \quad (\text{S.26})$$

5.1 Steady state - simplified system

Given the unknowns u, v^M, v^S, k, s , the following variables are determined sequentially:

$$r = \frac{1 - \beta}{\beta} + \delta \quad (\text{SS.1})$$

$$u^M = su \quad (\text{SS.2})$$

$$u^S = (1 - s)u \quad (\text{SS.3})$$

$$m^M = \mu_1 (v^M)^{\mu_2} (u^M)^{1-\mu_2} \quad (\text{SS.4})$$

$$m^S = \mu_1 (v^S)^{\mu_2} (u^S)^{1-\mu_2} \quad (\text{SS.5})$$

$$n^M = \frac{m^M}{\sigma^M} \quad (\text{SS.6})$$

$$n^S = \frac{m^S}{\sigma^S} \quad (\text{SS.7})$$

$$\psi^{hM} \equiv \frac{m^M}{u^M}, \quad (\text{SS.8})$$

$$\psi^{fM} \equiv \frac{m^M}{v^M}, \quad (\text{SS.9})$$

$$\psi^{hS} \equiv \frac{m^S}{u^S}, \quad (\text{SS.10})$$

$$\psi^{fS} \equiv \frac{m^S}{v^S} \quad (\text{SS.11})$$

$$l = 1 - u - n^M - n^S \quad (\text{SS.12})$$

$$S = B(n^S)^b \quad (\text{SS.13})$$

$$M = \left[\zeta k^{\frac{\alpha-1}{\alpha}} + (1 - \zeta)(n^M)^{\frac{\alpha-1}{\alpha}} \right]^{\frac{\alpha}{\alpha-1}}, \quad (\text{SS.14})$$

$$Y = \left[\gamma M^{\frac{\chi-1}{\chi}} + (1 - \gamma)S^{\frac{\chi-1}{\chi}} \right]^{\frac{\chi}{\chi-1}}, \quad (\text{SS.15})$$

$$p^S = (1 - \gamma) \left(\frac{Y}{S} \right)^{\frac{1}{\chi}} \quad (\text{SS.16})$$

$$p^M = \gamma \left(\frac{Y}{M} \right)^{\frac{1}{\chi}} \quad (\text{SS.17})$$

The wages of two sectors are obtained from the inverted FOCs with respect to sectoral employment:

$$w^S = p^S b \frac{S}{n^S} - \frac{\kappa^S}{\psi^{fS} \beta} + \frac{(1 - \sigma^S) \kappa^S}{\psi^{fS}} \quad (\text{SS.18})$$

$$w^M = p^M (1 - \zeta) \left(\frac{M}{n^M} \right)^{\frac{1}{\alpha}} - \frac{\kappa^M}{\psi^{fM} \beta} + \frac{(1 - \sigma^M) \kappa^M}{\psi^{fM}} \quad (\text{SS.19})$$

$$\bar{b} = \varpi \frac{(w^M n^M + w^S n^S)}{n^M + n^S} \quad (\text{SS.20})$$

$$T = \bar{b} u \quad (\text{SS.21})$$

$$c = k(r - \delta) + w^M n^M + w^S n^S + \bar{b} u - T + \Pi^M + \Pi^S \quad (\text{SS.22})$$

$$\lambda_c = c^{-\eta} \quad (\text{SS.23})$$

$$\lambda_{n^M} = \frac{\beta}{1 - \beta(1 - \sigma)} (c^{-\eta} w^M - \Phi l^{-\varphi}) \quad (\text{SS.24})$$

$$\lambda_{n^S} = \frac{\beta}{1 - \beta(1 - \sigma)} (c^{-\eta} w^S - \Phi l^{-\varphi}) \quad (\text{SS.25})$$

To solve for the unknowns, we numerically find the solution to the following system of equations, in which we use expressions (S.1) to (S.25).

$$\Phi l^{-\varphi} = \lambda_{n^M} \psi^{hM} s + \lambda_{n^S} \psi^{hS} (1 - s) + \lambda_c \bar{b} \quad (\text{SS.26})$$

$$p^M \cdot \zeta \left(\frac{M}{k} \right)^{1-\alpha} - r = 0 \quad (\text{SS.27})$$

$$\frac{\kappa^S}{\psi^{fS}} = \beta \left[p^S b \frac{S}{n^S} - w^S + \frac{(1 - \sigma^S) \kappa^S}{\psi^{fS}} \right] \quad (\text{SS.28})$$

$$\frac{\kappa^M}{\psi_t^{fM}} = \beta \left[p^M (1 - \zeta) \left(\frac{M}{n^M} \right)^{1-\alpha} - w^M + \frac{(1 - \sigma^M) \kappa^M}{\psi^{fM}} \right] \quad (\text{SS.29})$$

$$w^S = (1 - \vartheta^S) \left(p^S b \frac{S}{n^S} + \frac{(1 - \sigma^S) \kappa^S}{\psi^{fS}} \right) + \frac{\vartheta^S}{\lambda_c} (\Phi l^{-\varphi} - (1 - \sigma^S) \lambda_{n^S}) \quad (\text{SS.30})$$

$$w^M = (1 - \vartheta^M) \left(p^M (1 - \zeta) \left(\frac{M}{n^M} \right)^{1-\alpha} + \frac{(1 - \sigma^M) \kappa^M}{\psi^{fM}} \right) + \frac{\vartheta^M}{\lambda_c} (\Phi l^{-\varphi} - (1 - \sigma^M) \lambda_{n^M}) \quad (\text{SS.31})$$

$$\lambda_{n^M} \psi^{hM} = \lambda_{n^S} \psi^{hS} \quad (\text{SS.32})$$

6 The Role of the Frisch Elasticity of Labor Supply

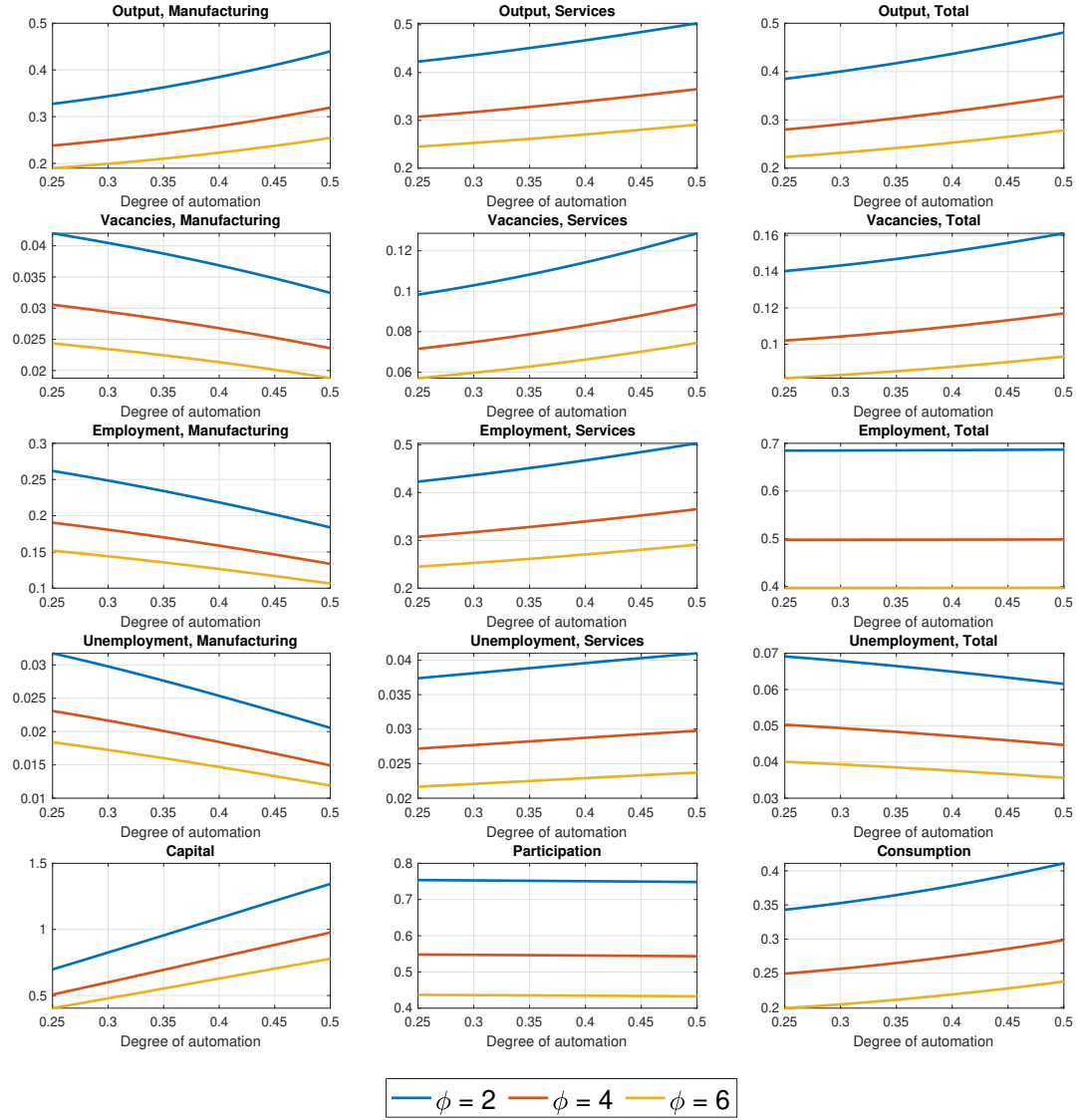


Figure A.1: Steady-state effects of automation in a two-sector economy: Different values of the Frisch elasticity

Note: The y-axis shows steady-state levels. The blue line refers to the baseline calibration ($\phi = 2$).

7 The Role of the Elasticities of Substitution

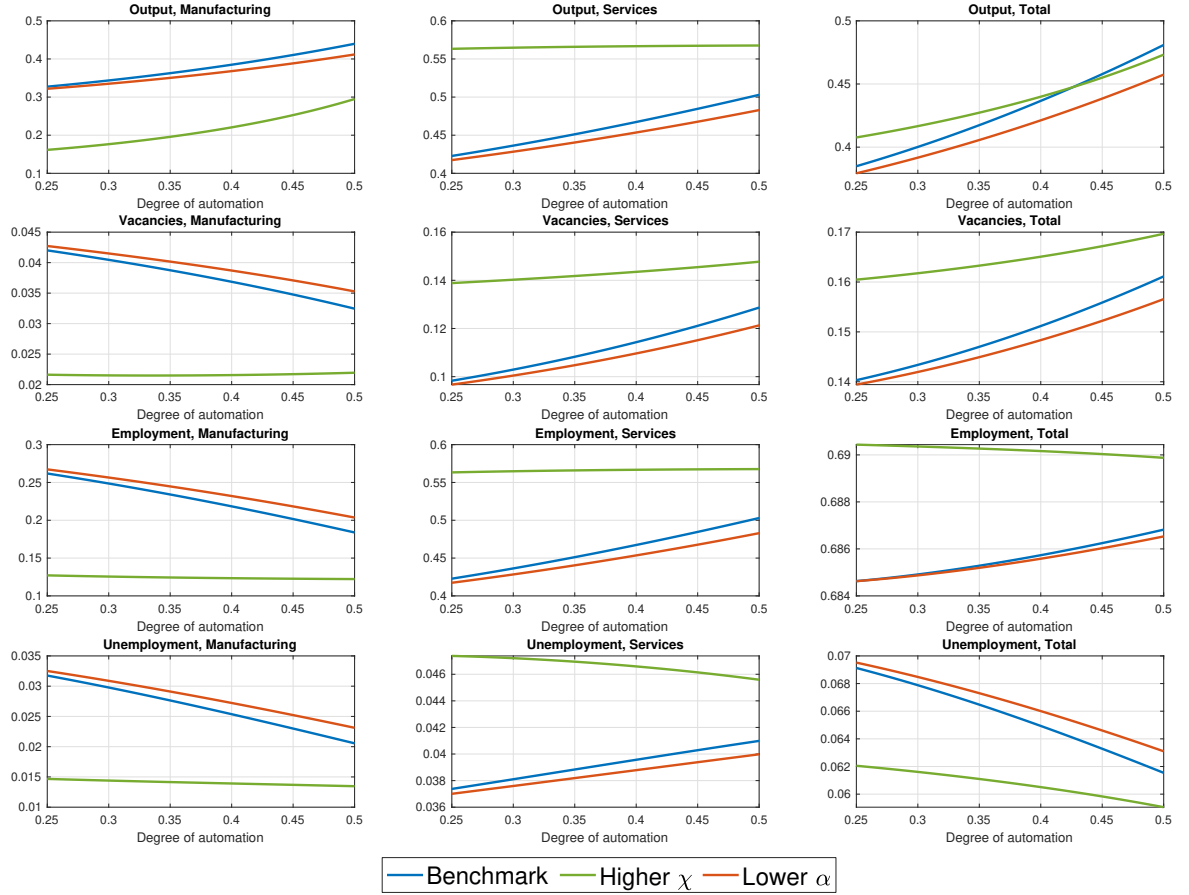


Figure A.2: Steady-state effects of automation in a two-sector economy: Different elasticities of substitution between capital and labor ($\alpha = 0.7$) and between the two goods ($\chi = 1.5$)

Note: The y-axis shows steady-state levels.

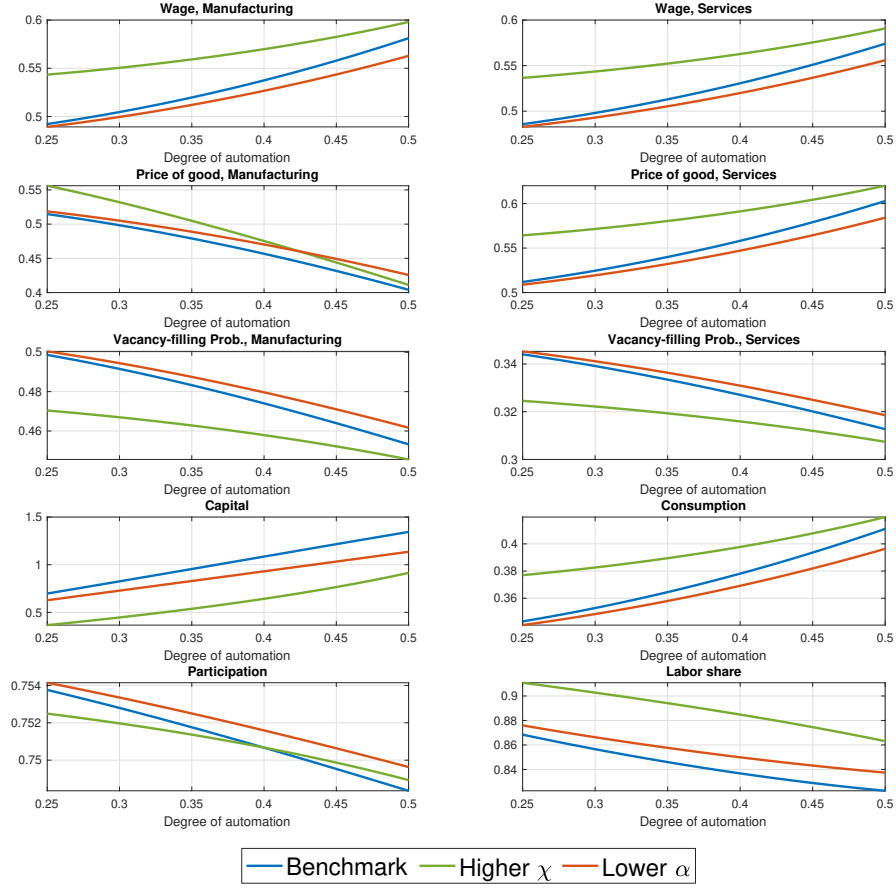


Figure A.3: Steady-state effects of automation in a two-sector economy: Different elasticities of substitution between capital and labor ($\alpha = 0.7$) and between the two goods ($\chi = 1.5$) (continued)

Note: The y-axis shows steady-state levels.

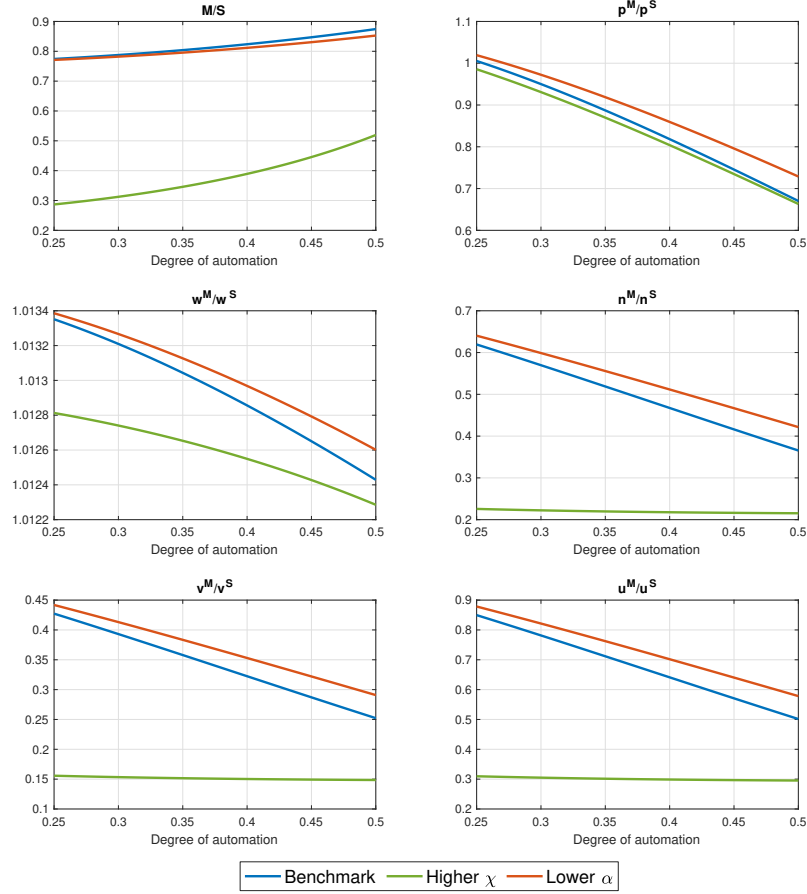


Figure A.4: Steady-state effects of automation on key ratios in a two-sector economy: Different elasticities of substitution between capital and labor ($\alpha = 0.7$) and between the two goods ($\chi = 1.5$)

Note: The y-axis shows steady-state levels.